

Factor (Mis)allocation :

A Review of the Literature

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Abstract

A central question of modern macro-development economics is why some economies produce so much less from the same measured inputs than others. One powerful answer is *misallocation*: even when each plant uses a good technology, aggregate output is lost if inputs are distributed across plants in the wrong proportions, so that an hour of labour or a rupee of capital yields a high return in one plant and a low return in another. This review develops the theory and the measurement of misallocation, organised around the framework of Hsieh & Klenow (2009). We begin with the conceptual foundations—what “misallocation” means, the idea of a *wedge*, and the two research traditions (the direct-distortion approach of Restuccia & Rogerson (2008) and the indirect, dispersion-based approach of Hsieh and Klenow). We then work through the Hsieh–Klenow model equation by equation: the demand and technology structure, the firm’s problem under output and capital wedges, the marginal-revenue-product conditions, the distinction between *physical* productivity (TFPQ) and *revenue* productivity (TFPR), and the central result that the dispersion of revenue productivity is—under the model’s assumptions—a sufficient statistic for the output loss from misallocation. For *each* equation we state its economic meaning, explain how it is taken to data, and say how the resulting estimate is read. We then show how the framework specialises to the labour market, where the dispersion of the marginal revenue product of labour (MRPL) becomes the object of interest and the wage *markdown* of Review A re-enters as the labour

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wedge—the bridge this thesis builds. A dedicated section sets out estimation in practice (data construction, the choice of the substitution elasticity and factor shares, dispersion and decomposition measures). We then survey the critiques that define the frontier: the measurement-error correction of Bils, Klenow, & Ruane (2021), the sufficient-statistic critique of Haltiwanger, Kulick, & Syverson (2018), the adjustment-cost critique of Asker, Collard-Wexler, & De Loecker (2014), and the variable-markup critique of Li & Wang (2021) and Baqaee & Farhi (2020). We close with the empirical record for the world and for India—including the thesis’s own finding that wage markdowns account for the better part of labour misallocation in Indian manufacturing—and with the open questions that Review C takes up.

Keywords: misallocation; total factor productivity; revenue productivity (TFPR); marginal revenue product of labour; Hsieh–Klenow; wage markdown; monopsony; Indian manufacturing.

JEL: O47, O11, D24, E23, L11, O14.

1 Introduction: from technology to allocation

Two plants in the same four-digit industry, making the same product with the same nominal technology, can nonetheless contribute very differently to aggregate output if one is starved of inputs and the other is over-supplied. If the under-resourced plant would earn a higher return on an extra worker or an extra machine than the over-resourced plant earns, then moving inputs from the second to the first raises total output *with no change in technology and no extra inputs in aggregate*. The size of the achievable free lunch is a measure of *misallocation*. The idea is old in spirit—it is the logic of the gains from trade applied inside a country—but it became a quantitative research programme with Restuccia & Rogerson (2008) and, above all, Hsieh & Klenow (2009), who showed how to measure it from ordinary manufacturing-census data and found that equalising the marginal products observed in the United States would raise manufacturing total factor productivity (TFP) by 30–50% in China and 40–60% in India.

This review matters to the thesis for a specific reason. Review A established that Indian manufacturing plants pay labour *below* its marginal revenue product—there is a wage *markdown*, $\psi_{it} > 1$. A markdown is, by definition, a wedge between what an extra worker produces in revenue and what the worker is paid; and a wedge that differs across plants is precisely the kind of distortion that generates misallocation. The thesis’s second objective is therefore to ask: *how much of the misallocation of labour in Indian manufacturing is due to dispersion in this wedge—to differences across plants in the degree of labour-market power—rather than to taxes, credit constraints, regulation, or measurement error?* To pose that question precisely we need the misallocation framework,

and to answer it credibly we need to understand the framework’s machinery and its critiques. That is the purpose of Review B.

The plan is as follows. Section 2 fixes concepts: the decomposition of aggregate TFP into a technology term and an allocation term, the language of wedges, and the two traditions of measurement. Section 3 develops the Hsieh–Klenow model equation by equation, with each equation accompanied by its economic meaning, the way it is estimated, and the way its estimate is interpreted. Section 4 specialises the framework to labour and shows where the markdown enters. Section 5 is a practical manual of estimation. Section 6 surveys the measurement alternatives and the major critiques. Section 7 reviews the evidence—global, Indian, and the thesis’s own. Section 8 states the debates and the gap, and hands over to Review C, which joins misallocation to the market-power theory of Review A.

2 Conceptual foundations: TFP, wedges, and two traditions

2.1 Aggregate productivity as technology plus allocation

The deepest reason to care about misallocation is that measured aggregate TFP is not a pure index of technology. Consider an economy of many plants i in a sector, each with physical productivity A_i (output per bundle of inputs) and each commanding a share s_i of the sector’s inputs. A useful and very general way to see the point is the decomposition popularised by Olley & Pakes (1996): writing $\omega_i = \ln A_i$ for log productivity and s_i for a plant’s input (or activity) share, sector log productivity can be written as

$$\Omega = \sum_i s_i \omega_i = \bar{\omega} + \sum_i (s_i - \bar{s})(\omega_i - \bar{\omega}) = \bar{\omega} + \text{Cov}(s_i, \omega_i), \quad (1)$$

where $\bar{\omega}$ is the unweighted mean of log productivity and $\bar{s} = 1/N$. The first term is *technology*: how productive the average plant is. The second term, the covariance between size and productivity, is *allocation*: it is positive when larger plants are the more productive ones (inputs are in the right hands) and small or negative when they are not. *Economic meaning.* Two economies with identical distributions of plant-level technology $\{A_i\}$ can have very different aggregate productivity if they differ in the allocation term $\text{Cov}(s_i, \omega_i)$. Development accounting that ignores allocation can mistake a badly organised economy for a technologically backward one.

How it is estimated. Equation (1) is computed directly: take a cross-section of plants, measure ω_i (a productivity index from a production-function estimate, or simply log labour productivity) and s_i (employment or value-added share), and form the sample covariance. Bartelsman, Haltiwanger, & Scarpetta (2013) do exactly this across countries and show that the allocation term is systematically larger (more positive) in the United

States than in many European and transition economies—a first-order cross-country productivity gap that is about allocation, not technology.

How it is interpreted. A rising covariance term over time signals improving allocative efficiency (reforms, entry/exit, the shrinking of protected laggards); a falling term signals deteriorating allocation. The limitation—which motivates the rest of this review—is that (1) tells us *that* allocation matters but not *why* inputs are mis-assigned. For that we need a model of the forces that pull plants away from their efficient size: wedges.

2.2 Wedges: the language of distortions

A *wedge* is a multiplicative gap inserted into a first-order condition that would, without it, equate a marginal product to a factor price. If a plant’s optimal choice of labour satisfies $\text{MRPL}_i = (1 + \tau_i^L)w$, then τ_i^L is the labour wedge: $\tau_i^L > 0$ means the plant behaves as if labour were more expensive than the market wage (it under-hires), $\tau_i^L < 0$ as if labour were cheaper (it over-hires). Wedges are a reduced-form bookkeeping device—they summarise the net effect of *whatever* distorts the plant’s choice (taxes, subsidies, credit constraints, adjustment costs, regulation, transport costs, market power) without committing to a particular cause. This “wedge accounting” descends from the business-cycle accounting of Chari, Kehoe, & McGrattan (2007) and is what makes the misallocation literature both powerful and contested: powerful because one can measure wedges without knowing their source, contested because a measured wedge is only as informative as the model that maps data into it.

The crucial economic fact is this: a wedge that is the *same* for every plant does not cause misallocation—it scales everyone equally and leaves relative allocations untouched. Misallocation is generated by the *dispersion* of wedges across plants. This is why the empirical literature studies the *variance* of marginal revenue products, not their level, and why Review A’s finding that the markdown *varies* across plants—not merely that it exceeds one—is what makes it a candidate source of misallocation.

2.3 Two traditions of measurement

The literature splits into two complementary traditions.

The *direct* or “distortion-first” tradition, exemplified by Restuccia & Rogerson (2008), starts from an explicit distribution of plant-level distortions (idiosyncratic taxes/subsidies on output or capital), feeds it through a general-equilibrium model of heterogeneous plants in the spirit of Hopenhayn & Rogerson (1993), and computes how much TFP would rise if the distortions were removed. Its strength is transparency about mechanisms and the ability to ask counterfactual policy questions (e.g. what a size-dependent tax costs). Its characteristic lesson is that distortions *correlated with productivity*—policies that, in effect, tax the most productive plants—are far more damaging than random

distortions of the same variance, because they actively prevent good plants from growing (Restuccia & Rogerson, 2017).

The *indirect* or “dispersion-first” tradition, due to Hsieh & Klenow (2009), runs the logic in reverse: rather than positing distortions and deriving dispersion, it *reads dispersion off the data* and infers the implied distortions and output loss. Its great advantage is that it requires only standard plant-level revenue and input data—no need to observe the policies themselves—which is why it has been applied to dozens of countries, India among them. Its great vulnerability, which Section 6 develops, is that anything other than a genuine distortion that also shows up as dispersion in revenue productivity—measurement error, demand shocks, adjustment costs, variable markups, technological heterogeneity—will be *mistaken* for misallocation. The thesis works squarely in this second tradition, and inherits both its reach and its hazards. We now build it up carefully.

3 The Hsieh–Klenow framework, equation by equation

The model has four ingredients: a demand system, a technology, a set of wedges, and an equilibrium concept. We take them in turn, stating each equation’s meaning, its route to the data, and the reading of its estimate.

3.1 Demand and technology

There is a single final good produced from the output of S sectors with a Cobb–Douglas aggregator, and each sector’s output is a constant-elasticity-of-substitution (CES) aggregate of the varieties produced by its M_s plants:

$$Y = \prod_{s=1}^S Y_s^{\theta_s}, \quad \sum_s \theta_s = 1, \quad Y_s = \left(\sum_{i=1}^{M_s} Y_{si}^{(\sigma-1)/\sigma} \right)^{\sigma/(\sigma-1)}. \quad (2)$$

Here $\sigma > 1$ is the elasticity of substitution between varieties within a sector and θ_s is the (Cobb–Douglas) expenditure weight of sector s .

Economic meaning. The CES nest says that within a sector, plants make imperfectly substitutable varieties; each plant therefore faces a downward-sloping demand curve with constant price-elasticity σ , and so charges a fixed gross markup $\sigma/(\sigma - 1)$ over marginal cost in the baseline model. The Cobb–Douglas top layer fixes each sector’s expenditure share at θ_s , which simplifies aggregation.

How it is taken to data. The CES demand curve implies that a plant’s revenue share within its sector is $P_{si}Y_{si}/(P_sY_s) = (P_{si}/P_s)^{1-\sigma}$, which is exactly what lets the analyst recover unobserved physical output from observed revenue (below). The parameter σ is *not* estimated inside the framework; it is imported—Hsieh and Klenow set $\sigma = 3$ as a baseline (a gross markup of 1.5) and report robustness at $\sigma = 4, 5$. The sector weights θ_s

are read from expenditure or value-added shares.

How it is interpreted. The choice of σ is consequential: it multiplies every estimated efficiency loss (Section 3.5). A larger σ means varieties are closer substitutes, so a given dispersion in revenue productivity implies larger quantity misallocation and a bigger output gain from reallocation. Any reported gain must therefore always be read *at a stated* σ .

Each plant has a Cobb–Douglas technology in capital and labour with *physical* (Hicks-neutral) productivity A_{si} :

$$Y_{si} = A_{si} K_{si}^{\alpha_s} L_{si}^{1-\alpha_s}, \quad 0 < \alpha_s < 1. \quad (3)$$

Economic meaning. A_{si} is *physical* total factor productivity—“TFPQ,” output per unit of input bundle—a pure technology object. The capital elasticity α_s is allowed to vary by sector but not across plants within a sector: a strong homogeneity assumption that the critiques in Section 6 attack.

How it is taken to data. α_s is set, not estimated plant-by-plant: Hsieh and Klenow use the U.S. industry cost share of capital (on the view that the U.S. is relatively undistorted, so cost shares reveal technology) times an adjustment, common to all plants in the sector. Crucially, physical output Y_{si} is *not observed*—censuses record sales (revenue) $P_{si}Y_{si}$, not quantities—so A_{si} cannot be computed directly; it is backed out using the demand curve, as the next subsection shows.

How it is interpreted. Because α_s is imposed from a (foreign, undistorted) benchmark, any genuine cross-plant variation in capital intensity that reflects technology rather than distortion will be misread as a wedge. Holding α_s common is the price paid for being able to identify the model from revenue data alone.

3.2 The firm’s problem and the wedges

Distortions enter as two plant-specific wedges: an *output* wedge τ_{si}^Y that taxes revenue (a stand-in for anything that scales the plant’s whole operation—size restrictions, output taxes, transport costs, demand frictions) and a *capital* wedge τ_{si}^K that raises the cost of capital relative to labour (credit constraints, investment subsidies, land-access frictions). The plant maximises profit

$$\pi_{si} = (1 - \tau_{si}^Y) P_{si}Y_{si} - wL_{si} - (1 + \tau_{si}^K) R K_{si}, \quad (4)$$

taking the wage w and rental rate R as common across plants within the economy.

Economic meaning. The deliberate asymmetry—a wedge on capital but not (in the baseline) on labour—reflects the original focus on capital-market frictions in developing economies. The labour wedge is normalised away in the canonical model; *re-introducing*

it is exactly what the thesis does (Section 4). The wedges are defined residually: they are whatever makes the plant’s observed choices optimal.

How it is taken to data. The wedges are not measured from outside information; they are inferred from the first-order conditions below, given the data on revenue, capital, labour and the imposed parameters. This is the defining move of the indirect tradition—and the reason a misspecified model produces spurious wedges.

How it is interpreted. A high τ_{si}^K for a plant says it acts as if capital were expensive (it is capital-starved relative to its productivity); a high τ_{si}^Y says the plant is held below its efficient scale overall. Only the *dispersion* of these wedges across plants matters for aggregate TFP.

3.3 First-order conditions: the marginal revenue products

Differentiating (4) and using the CES demand elasticity σ (so that marginal revenue is $(1 - 1/\sigma)$ times price) gives the two conditions at the heart of the framework. For labour and capital respectively,

$$\text{MRPL}_{si} \equiv \left(1 - \frac{1}{\sigma}\right) (1 - \alpha_s) \frac{P_{si} Y_{si}}{L_{si}} = \frac{w}{1 - \tau_{si}^Y}, \quad (5)$$

$$\text{MRPK}_{si} \equiv \left(1 - \frac{1}{\sigma}\right) \alpha_s \frac{P_{si} Y_{si}}{K_{si}} = \frac{(1 + \tau_{si}^K) R}{1 - \tau_{si}^Y}. \quad (6)$$

Economic meaning. The left-hand sides are *marginal revenue products*—the extra revenue from one more unit of the input, equal to (markup-adjusted) the input’s revenue share divided by its quantity. In a frictionless economy ($\tau^Y = \tau^K = 0$ for all plants) every plant faces the same w and R , so MRPL and MRPK are *equalised across plants*: that equality is what “no misallocation” means. Wedges break the equality. Note that MRPL depends only on the output wedge, while MRPK depends on both: capital bears the output wedge *and* its own capital wedge.

How it is taken to data. Both marginal revenue products are computable from observables alone: revenue $P_{si} Y_{si}$, the wage bill or employment L_{si} , the capital stock K_{si} , the imposed α_s and σ . One does *not* need to observe physical quantities or prices to compute MRPL and MRPK—this is the framework’s great practical advantage. The wedges are then read off: $1 - \tau_{si}^Y = w/\text{MRPL}_{si}$ and $1 + \tau_{si}^K = \text{MRPK}_{si}(1 - \tau_{si}^Y)/R$.

How it is interpreted. A plant with a high MRPK relative to peers is capital-constrained—giving it more capital would raise output a lot; a plant with a low MRPK is capital-glutted. The *cross-plant dispersion* of MRPK (and of MRPL) is the raw empirical signature of misallocation. The thesis’s entire second objective is the labour analogue: the dispersion of MRPL, and its decomposition into wage-setting power and other wedges.

3.4 Physical versus revenue productivity: TFPQ and TFPR

The framework’s pivotal conceptual move is to distinguish two productivity measures. *Physical* productivity (TFPQ) is A_{si} from (3). *Revenue* productivity (TFPR) is output value per input bundle:

$$\text{TFPR}_{si} \equiv P_{si}A_{si} = \frac{P_{si}Y_{si}}{K_{si}^{\alpha_s}L_{si}^{1-\alpha_s}} \propto (\text{MRPK}_{si})^{\alpha_s}(\text{MRPL}_{si})^{1-\alpha_s} \propto \frac{(1 + \tau_{si}^K)^{\alpha_s}}{1 - \tau_{si}^Y}. \quad (7)$$

Economic meaning. This is the single most important equation in the literature. In the *absence* of distortions, TFPR_{si} is *equalised across plants within a sector*: a more productive plant (higher A_{si}) optimally grows until its price P_{si} falls by just enough to leave the product $P_{si}A_{si}$ constant. So under the model, *any dispersion in TFPR is a fingerprint of distortions*, and the amount of dispersion measures the amount of misallocation. TFPQ varies across plants (plants differ in true technology); TFPR should not, unless wedges are present.

How it is taken to data. TFPR is directly computable from revenue, capital, labour and α_s —no prices needed—because the unobserved P_{si} and Y_{si} enter only as the product $P_{si}Y_{si}$, which *is* observed. TFPQ is recovered separately by inverting CES demand: combining the demand curve (which ties a plant’s relative price to its relative quantity) with (3) yields, up to a sector normalisation κ_s ,

$$A_{si} = \kappa_s \frac{(P_{si}Y_{si})^{\sigma/(\sigma-1)}}{K_{si}^{\alpha_s}L_{si}^{1-\alpha_s}}, \quad (8)$$

so physical productivity is backed out from *revenue raised to the power $\sigma/(\sigma - 1)$* . (This is why a plant with twice the revenue is inferred to have *more* than twice the physical output: high-revenue plants are inferred to have low prices.)

How it is interpreted. A plant with high TFPR is one whose marginal products are high—it is *constrained*, held below its efficient size; a plant with low TFPR is over-resourced. Note the counter-intuition that the empirical literature constantly flags: under the model, a “well-run” high-TFPR plant is one we should *expand*, because its high TFPR signals it is being held back, not that it is inefficient. This is also the hinge of the critiques: if high TFPR instead reflects a positive demand shock, a quality advantage, or measurement error, the model will wrongly label a thriving plant as “distorted.”

3.5 The sufficient statistic: TFPR dispersion and the efficiency gain

The payoff of the construction is a formula for the output achievable by reallocating inputs to equalise marginal products, holding technology $\{A_{si}\}$ and total inputs fixed. Aggregating the CES sector and comparing the distorted equilibrium to the efficient one,

sector TFP in the efficient allocation relative to the actual is

$$\frac{Y_s^{\text{efficient}}}{Y_s} = \left[\sum_{i=1}^{M_s} \left(\frac{A_{si}}{\bar{A}_s} \frac{\overline{\text{TFPR}}_s}{\text{TFPR}_{si}} \right)^{\sigma-1} \right]^{1/(\sigma-1)}, \quad (9)$$

where $\overline{\text{TFPR}}_s$ is a (harmonic, share-weighted) sector average. When $\ln \text{TFPR}$ is approximately normally distributed across plants, (9) collapses to the celebrated closed form

$$\ln \left(\frac{Y_s^{\text{efficient}}}{Y_s} \right) \approx \frac{\sigma}{2} \text{Var}(\ln \text{TFPR}_{si}). \quad (10)$$

Economic meaning. The output loss from misallocation is, to a close approximation, *half the substitution elasticity times the variance of log revenue productivity*. This is the “sufficient-statistic” result: one number, $\text{Var}(\ln \text{TFPR})$, summarises the whole loss. It is intuitive—more dispersion in TFPR means marginal products are further from being equalised, so more output is left on the table—and the factor σ appears because closer-substitute varieties make a given misallocation costlier.

How it is taken to data. Compute TFPR_{si} for every plant from (7); take logs; compute the within-sector variance; plug into (10) (or, better, evaluate the exact (9) by summing over plants). Aggregate across sectors with the Cobb–Douglas weights θ_s to get the economy-wide gain $Y^{\text{efficient}}/Y$. In practice the analyst (i) winsorises or trims TFPR tails to limit the influence of outliers and measurement error, (ii) demeans TFPR within industry×year so that only *within*-industry dispersion counts, and (iii) reports the gain at several values of σ .

How it is interpreted. The number is a *hypothetical upper bound*: the TFP gain if *all* within-industry gaps in marginal products were costlessly eliminated. It is not a forecast of any feasible reform; it is the size of the prize. A gain of “+100%” means manufacturing TFP could in principle double. Because the statistic scales with σ and with the variance, two researchers can report very different gains for the same economy purely through these choices—which is why the literature insists on reporting the underlying $\text{Var}(\ln \text{TFPR})$ itself, a parameter-free description of the data, alongside the headline gain. The thesis follows this discipline, reporting both the dispersion and the implied gain at $\sigma = 3, 4, 5$.

4 From capital and output wedges to the labour markdown

4.1 Re-instating the labour wedge

The canonical model of Section 3 normalises the labour wedge to zero: labour is bought competitively at a common wage w , and all the action is in the output and capital wedges. But Review A showed that this is false for Indian manufacturing: plants exercise *monopsony power* and pay a wage below the marginal revenue product of labour. Formally,

re-introduce a labour wedge so that the labour first-order condition (5) becomes

$$\text{MRPL}_{si} = \psi_{si} \frac{w_{si}}{1 - \tau_{si}^Y}, \quad \psi_{si} = 1 + \frac{1}{\varepsilon_{si}^L}, \quad (11)$$

where $\psi_{si} \geq 1$ is the wage *markdown* of Review A and ε_{si}^L is the firm-specific labour-supply elasticity. The labour wedge is now $1 + \tau_{si}^L = \psi_{si}$: a plant with more wage-setting power (lower ε^L) has a higher markdown and behaves as if labour were more expensive, hiring too few workers relative to the efficient benchmark.

Economic meaning. Equation (11) is the formal bridge of the thesis. It says the markdown is a *labour wedge*, and—like any wedge—it causes misallocation only insofar as it *differs across plants*. If every plant had the same markdown, labour would be mis-priced uniformly and relative allocations (hence aggregate TFP) would be untouched; it is the *dispersion* of ψ_{si} that misallocates labour.

How it is taken to data. MRPL_{si} is computed as in (5); the markdown ψ_{si} is estimated by the production approach of Review A (the Brooks et al. ratio $\psi = (\theta^L/\alpha^L)/(\theta^M/\alpha^M)$, using materials as a competitively-priced numeraire); the wage w_{si} is the observed wage bill per worker. With these in hand the labour wedge is identified plant-by-plant.

How it is interpreted. The object of interest becomes $\text{Var}(\ln \text{MRPL}_{si})$ —the labour analogue of $\text{Var}(\ln \text{TFPR})$ —and the question is how much of it is accounted for by the dispersion of the markdown $\ln \psi_{si}$ versus the dispersion of wages and other wedges. That decomposition is the empirical core of the thesis’s second objective.

4.2 Decomposing the dispersion of the marginal revenue product of labour

Taking logs of (11) (and folding the output wedge into the residual), the marginal revenue product of labour decomposes additively,

$$\ln \text{MRPL}_{si} = \ln \psi_{si} + \ln w_{si} + u_{si}, \quad (12)$$

so that its cross-plant variance splits into the variance of the markdown, the variance of wages, the variance of the residual, and their covariances.

Economic meaning. This is the accounting identity that lets the thesis answer “how much of labour misallocation is market power?”. The markdown term captures *factor-market power*; the wage term captures compensating differentials, skill, and local labour-market conditions; the residual captures output/credit wedges and noise.

How it is taken to data. Compute each component plant-by-plant, then form the variance decomposition $\text{Var}(\ln \text{MRPL}) = \text{Cov}(\ln \text{MRPL}, \ln \psi) + \text{Cov}(\ln \text{MRPL}, \ln w) + \text{Cov}(\ln \text{MRPL}, u)$; the share attributable to the markdown is $\text{Cov}(\ln \text{MRPL}, \ln \psi)/\text{Var}(\ln \text{MRPL})$. (Using covariance shares rather than raw variance shares makes the components add exactly to one even when they are correlated.)

How it is interpreted. A markdown share near one means labour misallocation *is* the dispersion of labour-market power; a share near zero means market power is incidental and other wedges dominate. The thesis’s headline result, set out in Section 7, is that for Indian organised manufacturing the markdown share is large—around four-fifths of the variance of $\ln \text{MRPL}$ —making labour-market power the principal source of the misallocation of labour, the empirical claim that motivates Review C.

5 Taking the framework to data: estimation in practice

This section is a practical manual: the steps an analyst actually performs, and the choices that drive the results.

1. Data and the unit of analysis. The framework needs plant- or firm-level revenue, employment (or the wage bill), capital, and—for the gross-output and markdown extensions—materials, all by industry and year. For India the natural source is the Annual Survey of Industries (ASI), the universe of registered factories above the employment thresholds, which the thesis uses for 2012–13 to 2023–24. The cross-sectional unit is the factory; industries are defined at the NIC two- to four-digit level; and all dispersion is measured *within* industry $\times$ year so that genuine technological differences across industries are not counted as misallocation.

2. Constructing the marginal products and TFPR. Real revenue is gross value of output (or value added) deflated by industry price indices; the capital stock is built by perpetual inventory or taken from book values adjusted to replacement cost; labour is total persons engaged or the wage bill. With α_s set from cost shares and σ imposed, MRPL_{si} , MRPK_{si} and TFPR_{si} follow mechanically from (5)–(7). The most consequential data choices are the capital measure (the noisiest input, and the one most contaminated by measurement error—see Bills, Klenow, & Ruane (2021)) and the deflators.

3. Choosing the elasticity of substitution and factor shares. σ is imported, not estimated: the convention is $\sigma = 3$ with robustness at 4 and 5. Factor shares α_s are taken from an undistorted benchmark (U.S. shares, or the sample’s own cost shares if one is willing to assume the domestic factor markets are competitive—an assumption the markdown literature explicitly rejects, which is itself a reason to prefer the gross-output/markdown route). Reporting the gain at several (σ, α_s) settings is mandatory because, by (10), the headline gain is linear in σ .

4. Measuring dispersion. The parameter-free description of the data is the *standard deviation of log TFPR* (and of $\log \text{MRPL}$, $\log \text{MRPK}$); the *90–10* and *75–25* log ratios

are reported as robust, tail-insensitive companions. These should be shown before any efficiency-gain number, because the gain is just a σ -scaled transform of the variance. Trimming or winsorising the top and bottom 1%–2% of the TFPR distribution is standard to bound the influence of outliers and coding errors.

5. Computing the efficiency gain. Evaluate the exact sectoral ratio (9) plant-by-plant (preferred) rather than only the lognormal approximation (10), then aggregate with the sector weights θ_s . Report the *canonical value-added* gain (treating capital and labour as the only inputs) and, as a conservative lower bound, the *integrated gross-output* gain (treating materials as a freely-adjusted input that absorbs part of the distortion); the latter is smaller because flexible materials soak up some of the mismeasured wedge.

6. Decompositions. Two are standard. The *between/within* split attributes dispersion to differences across industries versus within them (only the within part is misallocation in the strict sense). The *Olley–Pakes covariance* (1) attributes the *level* of aggregate productivity to the average plant versus the size–productivity allocation. The thesis adds a third, novel to the labour reading: the variance decomposition (12) of log MRPL into markdown, wage and residual.

7. Trends and robustness. Because the headline gain is sensitive to modelling, the credible deliverable is a *trend*—is misallocation rising or falling?—together with the underlying dispersion series, shown to be robust across σ , across deflators, across the value-added/gross-output choice, and to measurement-error corrections. A trend is far more robust than a level, because many of the contaminants (fixed measurement error, time-invariant technological heterogeneity) difference out.

6 Measurement alternatives and the critiques that define the frontier

The sufficient-statistic logic is only as good as its assumptions. Five lines of critique—each with a constructive fix—define the current frontier, and each bears directly on how much of the thesis’s measured labour misallocation can be trusted.

6.1 Allocative efficiency without the full model: the covariance approach

Before the Hsieh–Klenow machinery, the Olley & Pakes (1996) covariance (1) already measured allocative efficiency, and it remains a valuable robustness check because it needs no demand system and no σ . Bartelsman, Haltiwanger, & Scarpetta (2013) show cross-country that the covariance term is the larger part of the U.S. productivity advantage. *How to estimate:* compute the size-weighted mean productivity minus the unweighted

mean. *How to interpret:* a positive, rising covariance is improving allocation; the measure is conservative (it captures the level of allocative efficiency, not the counterfactual gain) and immune to the σ -scaling controversy, so agreement between the covariance trend and the TFPR-dispersion trend is reassuring.

6.2 Measurement error: “misallocation or mismeasurement?”

The sharpest empirical critique is that revenue and inputs are measured with error, and classical measurement error *inflates* the variance of TFPR mechanically—error in the denominator of (7) masquerades as a distortion. Bills, Klenow, & Ruane (2021) propose a correction that exploits a testable implication: if a plant’s high measured average product is real (a true wedge), its revenue should respond one-for-one to input growth; if it is measurement error, revenue growth should be *less* sensitive to measured input growth. Estimating that sensitivity lets them strip out the error.

How it is estimated. Regress revenue growth on measured input growth; the slope’s departure from one identifies the share of TFPR dispersion that is noise; scale the implied gains accordingly. *What they find.* For Indian manufacturing the correction lowers the potential reallocation gains by about 20%; for the United States it is more dramatic, cutting the gains by 60% and eliminating two-thirds of an otherwise mysterious downward trend in allocative efficiency over 1978–2013. Earlier drafts put it starkly: true marginal products in Indian manufacturing were only about *half* as dispersed as measured average products. *How to interpret.* A large but not total share of measured Indian misallocation survives the correction—misallocation is real but its raw magnitude is overstated. The lesson for the thesis is to lean on *trends* and on measurement-error-robust comparisons, and to treat level estimates of the gain as upper bounds.

6.3 Misspecification: is the variance of TFPR even a sufficient statistic?

Haltiwanger, Kulick, & Syverson (2018)—in a paper pointedly titled “the distortion that ate the residual”—show that the sufficient-statistic result holds only under strict demand- and supply-side assumptions. Where they have plant-level prices and quantities to test the assumptions directly, the assumptions fail: much of the variation in revenue productivity reflects *demand* shifts (a plant with a hot product has high TFPR but is not distorted) rather than inefficiency. Under general demand and supply structures, they prove $\text{Var}(\ln \text{TFPR})$ is *not* a sufficient statistic for the efficiency loss, and worse, the framework is biased towards labelling *better* businesses as “distorted.” *How to interpret for the thesis:* this is why isolating a *specific, theory-disciplined* wedge—the labour markdown, identified from cost minimisation and an internal materials numeraire as in Review A—is more defensible than reading raw TFPR dispersion as “the” distortion. The markdown is a structural object with a clear behavioural meaning, not a residual

that absorbs every unmodelled force.

6.4 Dynamics and adjustment costs

Asker, Collard-Wexler, & De Loecker (2014) make a complementary point: capital is a *dynamic* input subject to adjustment costs, so even in a *frictionless, undistorted* economy a plant hit by a productivity shock cannot instantly resize, and its marginal product temporarily diverges from the common level. Much measured dispersion in MRPK is therefore the footprint of ordinary investment dynamics, not policy distortion. *How to estimate the distinction:* simulate a model with adjustment costs and no distortions, and show it reproduces a large share of the observed dispersion; what it cannot reproduce is the *persistent* component. *How to interpret:* the *transitory* part of marginal-product dispersion should be discounted; the *persistent* part is the better candidate for genuine misallocation. This favours the labour markdown again, since wage-setting power is a persistent, structural feature of a plant’s local labour market rather than a transitory shock.

6.5 Variable markups and technological heterogeneity

Finally, the baseline framework assumes a *constant* markup $\sigma/(\sigma - 1)$ and a *common* technology α_s . Relax either and TFPR dispersion ceases to be a clean distortion measure. Li & Wang (2021) show that with *variable* markups (incomplete cost pass-through), TFPR dispersion is no longer a sufficient statistic for misallocation, and that correcting for it makes China’s measured misallocation *considerably smaller*. Baqaee & Farhi (2020), in general equilibrium, show that markup dispersion is itself a distinct channel of aggregate TFP loss—on their estimates, markup misallocation costs on the order of 15% of TFP—so the markup is not a nuisance to be purged but a mechanism to be modelled. *How to interpret for the thesis:* variable markups and markdowns are not contaminants to be assumed away but the *very object* of study. This is precisely why Review C is needed: once market power is variable, the misallocation framework and the market-power framework are no longer separable, and the dispersion of MRPL must be read jointly with the dispersion of markups and markdowns. The thesis’s integrated gross-output measure—which nets out the product-market markup using materials as a competitive numeraire—is one disciplined response to this critique.

6.6 Taking stock of the critiques

The critiques do not demolish the framework; they discipline it. Their collective lesson is fourfold: (i) report dispersion, not just σ -scaled gains; (ii) prefer trends to levels; (iii) correct for measurement error or at least bound it; and (iv) anchor the analysis on a *structural* wedge with a behavioural interpretation rather than on the raw TFPR residual.

The thesis’s labour-markdown reading of misallocation satisfies all four, which is the methodological case for it.

7 The empirical record: world, India, and the thesis

7.1 The global picture

The founding estimates are those of Hsieh & Klenow (2009): equalising marginal products to the U.S. benchmark would raise manufacturing TFP by 30–50% in China and 40–60% in India, with the 90–10 ratio of plant TFPR around five-fold in India against roughly three-fold in the United States; fully equalising within-country TFPR (the upper-bound counterfactual) implies gains of the order of 100–130% for India and 85–115% for China, against 30–45% for the United States. The framework travelled widely: Brandt, Tombe, & Zhu (2013) find factor-market distortions reduced Chinese aggregate TFP by about 30%; Gopinath et al. (2017) document *rising* capital misallocation in southern Europe as cheap post-euro capital flowed to low-productivity firms; Bun et al. (2022) estimate misallocation cost the Netherlands around 14 percentage points of productivity, with capital misallocation the more permanent component. Against these stand the deflationary results: Bils, Klenow, & Ruane (2021) attribute a large share of measured dispersion to measurement error; Haltiwanger, Kulick, & Syverson (2018) attribute much of it to demand; Asker, Collard-Wexler, & De Loecker (2014) attribute much of the capital part to adjustment dynamics; and nonparametric, design-based estimates that exploit exogenous demand shocks find efficiency losses an order of magnitude *smaller* than parametric Hsieh–Klenow numbers (Carrillo et al., 2023). The honest reading is that misallocation is real and economically large, but its precise magnitude is model-dependent and its raw levels are overstated.

7.2 India

For India specifically, beyond the founding Hsieh–Klenow estimates, Kabiraj (2020) extends the framework to gross output using both formal and informal firms and—decomposing the variance of TFPR into output, capital, raw-material, energy and service-sector distortions—finds that *raw-material and output distortions* play the major role in defining aggregate misallocation, a result that foreshadows the thesis’s emphasis on wedges other than the capital wedge of the canonical model. The broad Indian narrative is of substantial within-industry dispersion in marginal products, only partly reduced by the post-1991 reforms, with the labour market a recurring suspect: rigid labour regulation, small-scale reservation, and—the thesis’s contribution—pervasive employer wage-setting power.

7.3 The thesis’s own results

Applying the framework to ASI 2012–13 through 2023–24, the thesis documents large but *declining* labour misallocation and traces the bulk of it to the wage markdown. The standard deviation of log MRPL is about 1.20 and of log MRPK about 1.45; the within-industry standard deviation of log TFPR falls from 0.410 in 2012–13 to 0.346 in 2023–24 (a 16% reduction), and the standard deviation of the log markdown is about 1.22, also drifting down. In the canonical Hsieh–Klenow value-added accounting the implied aggregate efficiency gain is +102% at $\sigma = 3$ (rising to +139% and +169% at $\sigma = 4, 5$)—squarely inside the 100–128% range Hsieh & Klenow (2009) report for India, an external validation of the implementation; the conservative integrated gross-output measure gives +16% at $\sigma = 3$. Under both measures the potential gain *falls* over the decade, consistent with the declining dispersion.

The thesis’s distinctive result is the decomposition (12). Across plants the *markdown accounts for about 79% of the variance of log MRPL*, wage dispersion for about 30%, and the net market-power channel (markdown plus product-market markup) for about 65%; equalising the markdown alone would cut the variance of log MRPL by about 56% (its standard deviation falling from 1.20 to 0.80). The decomposition is stable over time and across sample cuts (the markdown share stays in a narrow 0.75–0.84 band), and the cross-industry correlation between labour misallocation (SD log MRPL) and markdown dispersion (SD log ψ) is 0.83—far above the corresponding correlation with markup dispersion. In words: *the misallocation of labour in Indian organised manufacturing is, first and foremost, the dispersion of labour-market power*. That empirical claim is what makes the linkage of Review C not a theoretical curiosity but the central fact of the data.

8 Debates, open questions, and the bridge to Review C

Three debates remain open and frame the rest of the thesis.

First, how much misallocation is real? The measurement-error, demand, and adjustment-cost critiques each subtract from the raw Hsieh–Klenow gains, and the design-based estimates subtract a great deal. The defensible posture, which the thesis adopts, is to emphasise robust *trends* and a *structural* wedge (the markdown) rather than headline level gains.

Second, what is the source of the wedges? The canonical framework is agnostic; the direct tradition models specific policies. The thesis takes a stand for the labour market: the dominant wedge is wage-setting power, not the capital-market friction of the original model. This is both an empirical finding (the 79% markdown share) and a research programme—explaining *why* markdown power varies across plants is the task of the determinants analysis in Objective 3 and Review D.

Third, can market power and misallocation be measured separately at all? The variable-markup critique says no: once markups and markdowns vary across plants, the product-market and factor-market distortions interact, and the dispersion of MRPL cannot be read in isolation from the dispersion of μ and ψ . This is the precise point at which Review B hands over to Review C. Review A measured market power; Review B has shown that the dispersion of market power *is* a—indeed the leading—source of the misallocation of labour in Indian manufacturing. Review C develops the theory that joins the two: variable markups and markdowns under alternative demand systems (CES, Kimball, translog, oligopoly), the formal models in which market power generates misallocation, and the conditions under which equalising the markdown would deliver the efficiency gains this review has quantified. Review D then turns to the macroeconomic and trade determinants that move these wedges over time.

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